TELIPORT
Script Your Future
Season 4

TATA
TATA ELXSI

Case Study 2

Team Details

THINK. BUILD. CREATE. WIN

TELIPORT
Script Your Future
Season 4

Test a battery before you build it

VOLT-FORGE connects every battery design decision to what it physically changes

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PROBLEM UNDERSTANDING

One change moves everything

A battery design is scattered across datasheets, circuit tools, CAD and spreadsheets. Swap one copper link and current sharing, heat and cost all shift, and you usually find out on the bench. So we took one real job: keep a Raspberry Pi 5 gateway alive for **10 minutes at 25 W**.

DESIRABILITY

Break it here, not on the bench

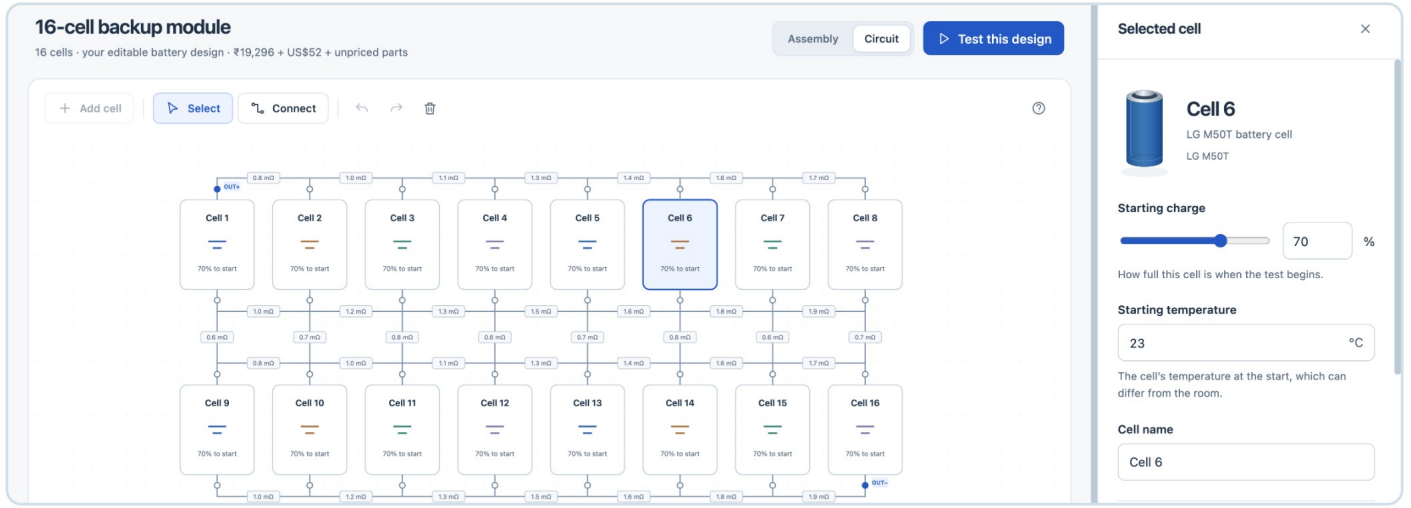
Students can't afford to burn parts just to learn what a bad connection does. Engineers can't afford to learn it late. Here you wire a real circuit, change one thing and see exactly what that decision changes, **before spending a rupee**.

TECHNICAL FEASIBILITY

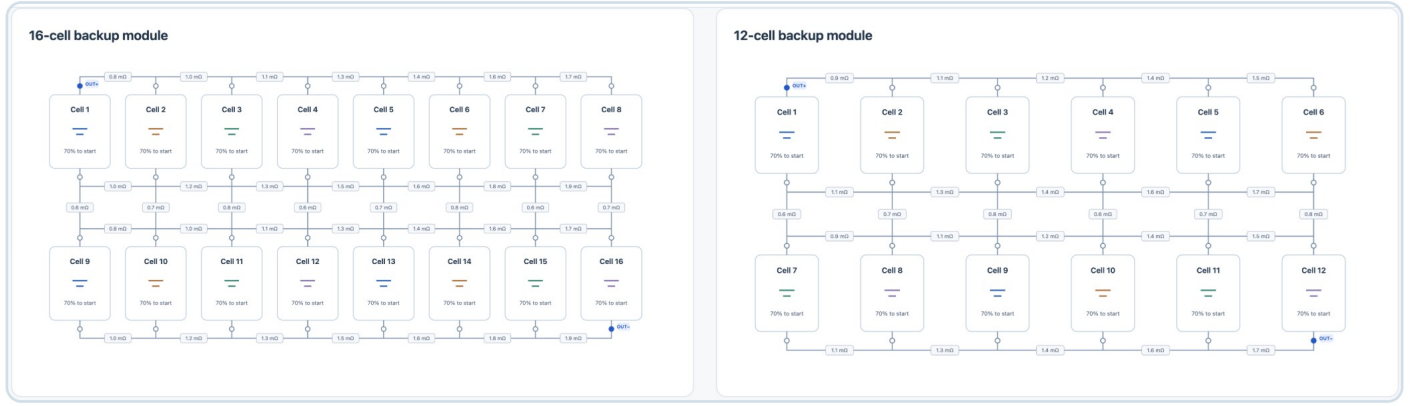
Runs on a laptop, not in a lab

Published LG M50T cell data and a local app. For the Pi job, 2 cells cleared **all 33 required cases**, with the busiest cell at 5.33 A against a 7.275 A limit, even with a mismatched cell in any position.

PROTOTYPE & EVIDENCE



Build. A real 16-cell circuit with 36 links. Click any cell and its properties open on the side.



Compare. Same 16 A load: 12 cells save ₹4,824 and 280 g, while the busiest cell works 0.25 A harder.

4,006 what-if runs. Weak cells, bad links, poor cooling: we tried to break it first.

15 of 15 PyBaMM checks. The open-source battery simulator matched our voltage and heat.

5 of 5 replays. Export a result and anyone can rerun it to the same numbers.

ENGINEERING QUALITY

The AI suggests, the solver proves

Every number comes from a circuit and heat solver, never a chatbot, and each run checks that heat matches electrical loss. Real-cell data showed cooling changes from rig to rig, so temperature stays a **range**, not a promise.

INNOVATION

It finds the answer for you

Testing what you drew is the easy part. **Find the smallest design** rebuilds the circuit at every size, tests each one and shows what survives. Our Pi backup went from 4 cells to 2, at half the cell cost and mass.

BUSINESS VIABILITY



41.3 lakh TAM, engineering students

15.6 lakh SAM, electrical, electronics, mechanical

130 SOM, colleges, 5% assumed

₹35 crore

to build all of it, from a measured dataset to the full app: 100 person-years, about 2% of AspenTech's yearly R&D. Free for students, paid for labs.

Everything here is simulated. Next we bench-test real cells and put it in front of students and engineers.